Flatten or Pool? Revisiting VGG16 Transfer Learning for Brain Tumor MRI Classification

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***Abstract*— Deep learning models for classifying brain tumors in magnetic resonance imaging (MRI) routinely report 96–99.6% accuracy on public benchmarks. We reproduce one such model, a VGG16 transfer-learning classifier reported by Wong et al. to reach 99.24% accuracy on four classes (glioma, meningioma, pituitary tumor, no tumor). Our reproduction reaches 95.25% test accuracy (95.38% in a second, independent run); three classes match the original within about one point, while glioma recall is lower (84.5% versus 100%). A duplicate audit shows that 202 of the 800 test images (25%) share an image fingerprint with a training image; accuracy is 99.50% on these and 93.81% on the remaining unseen images. Training curves show strong overfitting, which we attributed to the 6.4 million weights that the Flatten layer feeds into the classifier. We therefore tested one hypothesis, with predictions stated in advance: replacing Flatten with global average pooling (GAP), which cuts these weights 49-fold, reduces overfitting and improves accuracy. The hypothesis was refuted: GAP lowered test accuracy from 95.38% to 92.25% and glioma recall from 85.0% to 72.5%. The training curves explain why. GAP nearly halved validation loss while the backbone was frozen, but once VGG16 was fine-tuned both models overfit to a similar degree, indicating that the dominant source of overfitting is the fine-tuning stage rather than the classifier head. We discuss the resulting follow-up hypothesis and make no clinical claims.**

***Index Terms*— brain tumor, MRI, transfer learning, VGG16, global average pooling, reproducibility, data leakage**

I. Introduction

MRI is the preferred imaging modality for detecting brain tumors, but reading scans manually is slow and depends on the radiologist's experience. Convolutional neural networks (CNNs) promise fast, consistent screening, and a large body of work now classifies MRI slices into tumor types automatically.

Most of this work reports remarkably high accuracy. Across the ten studies we reviewed, very different methods (pretrained CNNs, custom residual networks, vision transformers and object detectors) mostly report 96–99.6% accuracy on four-class benchmarks. When accuracy has reached a ceiling, the more useful questions are whether the reported numbers can be reproduced, how they were measured, and which errors remain.

We study these questions using the model of Wong et al. [1], who fine-tune an ImageNet-pretrained VGG16 [14] on 7,023 MRI images and report 99.24% accuracy. No code was released, so we rebuild the method from its description. Our contributions are:

1. an independent reproduction of the base model, with every unspecified setting documented;
2. a duplicate audit showing that a quarter of the test images match training images, and measuring how much this inflates accuracy;
3. a single-variable experiment, replacing Flatten with GAP [16], whose predictions were written down before it was run, together with a second, independent baseline run;
4. a failure analysis using confusion matrices, misclassified examples and Grad-CAM [17].

II. Literature Review

We searched Google Scholar, arXiv and PubMed Central, found about 60 candidate papers, kept 24 that used this dataset or a close equivalent, and selected 10 that cover five method families (Table I).

1) Transfer learning. Disci et al. [2] compared six ImageNet-pretrained CNNs on the four-class dataset, with Xception performing best at 98.73%. Wong et al. [1], our base paper, fine-tuned VGG16 in two stages and tripled the training set with augmentation, reporting 99.24%.

2) Custom architectures. Zahoor et al. [3] designed a residual CNN (Res-BRNet) that copes well with class imbalance (98.22%). Srinivasan et al. [4] showed that detecting a tumor is much easier than naming its type: 99.5% for detection but 93.81% for five-type classification. Albalawi et al. [5] predicted tumor type and location with one multi-task model (about 99%) and discussed federated learning for privacy.

3) Explainability. Yoon and Lin [6] reproduced VGG16, ResNet50 and EfficientNetB0 (96–98%) and found that glioma and meningioma remain frequently confused. Musthafa et al. [7] paired ResNet50 with Grad-CAM (98.52%), but judged the heatmaps only by visual inspection.

4) Vision transformers. Reddy et al. [8] showed on the same dataset that a vision transformer outperforms ResNet50, EfficientNet-B0 and MobileNet-V2 baselines (94.9–96.5%).

5) Detection and surveys. Taha et al. [9] used YOLO object detectors, which also localize the tumor (YOLOv11: 99.56%; YOLOv8: 99.49%). Bouhafra and El Bahi [10] reviewed 60 studies from 2020–2024 and confirmed these trends across the field.

Evaluation practice. Work outside this set warns that random image-level splits can place near-identical slices of one patient in both training and test sets: Badža and Barjaktarović [12] report lower accuracy under subject-wise than record-wise

TABLE I
SUMMARY OF THE REVIEWED STUDIES

Study	Approach	Reported
Wong et al. [1] (base)	VGG16 + augmentation	99.24%
Disci et al. [2]	Xception (best of 6 CNNs)	98.73%
Zahoor et al. [3]	Custom residual CNN	98.22%
Srinivasan et al. [4]	Custom CNN, 5 types	93.81%
Albalawi et al. [5]	Multi-task CNN	∼99%
Yoon & Lin [6]	Reproduction, 3 CNNs	96–98%
Musthafa et al. [7]	ResNet50 + Grad-CAM	98.52%
Reddy et al. [8]	Vision transformer	best of 4
Taha et al. [9]	YOLOv11 detection	99.56%
Bouhafra & El Bahi [10]	Survey, 60 studies	–

cross-validation, and Yagis et al. [13] show that slice-level leakage substantially inflates brain MRI results.

III. Research Gaps

A. Gaps Across the Literature

G1: Explainability is shown, not measured. Heatmaps are inspected by eye; no reviewed study quantifies how well they overlap the tumor.

G2: Glioma–meningioma confusion persists. CNNs, residual networks and transformers all still confuse these types, suggesting the cause lies beyond the choice of architecture.

G3: Accuracy has reached a ceiling, while evaluation receives little scrutiny. Most reported accuracies cluster within about 3.5 points of each other (96–99.6%), yet leakage between training and test images is rarely checked.

B. Gaps in the Base Paper

G4: Signs of overfitting. The Flatten layer passes 25,088 values to a 256-unit layer, giving the classifier 6.4 million weights for 5,712 real training images; the published training curves show validation loss well above training loss.

G5: Unaudited test set. The data are split by image rather than by patient, and duplicate images between splits are not checked.

G6: No ablation and limited reporting. Four factors are credited for the accuracy but none is tested in isolation, results come from a single run, and AUC is not reported.

This study addresses **G4** directly through its hypothesis, measures **G5** through a duplicate audit, and reports per-class errors to examine **G2**.

IV. Hypothesis and Predictions

VGG16 turns each image into a $7 \times 7 \times 512$ block of feature maps; each of the 512 maps records where one visual pattern occurs. The classifier head must first turn this block into a vector. Fig. 1 shows the two options compared in this study.

Flatten (baseline) keeps every value, so the next layer receives 25,088 inputs and needs

$$25{,}088 \times 256 + 256 = 6{,}422{,}784 \text{ weights.} \quad (1)$$

Global average pooling averages each map into a single number, so the next layer receives 512 inputs and needs

$$512 \times 256 + 256 = 131{,}328 \text{ weights,} \quad (2)$$

about 49× fewer. Flatten records *where* each pattern occurs; GAP records only *how strongly* it occurs.

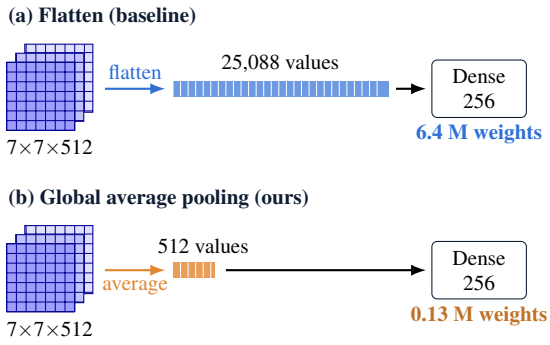


Fig. 1. The single change tested in this study. (a) Flatten keeps all 49 positions of every feature map, describing the image location by location. (b) GAP keeps one average per map, summarizing how strongly each pattern appears, and reduces the classifier's weights about 49×.

TABLE II
PREDICTIONS WRITTEN BEFORE THE EXPERIMENT

Measure	Prediction	Reason
Test accuracy	+1 to +3 pts	less memorization
Validation loss	30–60% lower	fewer confident errors
Train–val loss gap	smaller	direct overfitting sign
Glioma recall	higher	weakest class
Pituitary recall	may decrease	position cue lost
Macro AUC	slightly higher	better ranking

Hypothesis. *Replacing the Flatten layer with global average pooling will reduce overfitting, because a classifier with 49× fewer weights cannot easily memorize roughly 5,600 real training images and must rely on general tumor features instead.*

Following good experimental practice, we wrote down our predictions before running the experiment (Table II). We also anticipated a risk: pituitary tumors appear in a consistent anatomical location, so the baseline may use position as a cue that GAP discards, which could lower pituitary recall.

V. Methodology

A. Dataset

The base paper combines three public sources (Figshare [11], SARTAJ and Br35H), a combination also distributed on Kaggle [20]. The Kaggle copy we used turned out to be a different, balanced version (Table III). To match the base paper's validation and test sizes, the original test folder is split equally into validation and test sets for each class.

B. Pipeline

Fig. 2 summarizes the pipeline. Images are converted to grayscale, resized to 224×224, replicated to three channels and rescaled to $[0, 1]$. Each training image receives two augmented

TABLE III
DATA USED BY THE BASE PAPER AND BY THIS STUDY

	Wong et al. [1]	Ours
Total images	7,023	7,200
Training (real / after augmentation)	5,712 / 17,136	5,600 / 16,800
Validation / test	656 / 655	800 / 800
Class balance	unbalanced	25% each

TABLE IV
TRAINING SETTINGS (IDENTICAL FOR BOTH MODELS)

Setting	Stage 1	Stage 2
Trainable	head only	head + blocks 4–5
Optimizer, learning rate	Adam [19], 10^{-3}	Adam, 10^{-4}
Epochs	50	50
Batch size, steps/epoch	32, 179	32, 179
Loss	categorical cross-entropy	
Seed, hardware	42; NVIDIA T4 (Kaggle), TensorFlow 2.20	

copies (shear and zoom range 0.2, horizontal flip), tripling the training set; validation and test images are not augmented. The VGG16 base [14], pretrained on ImageNet [15], is followed by the pooling layer under study, a 256-unit ReLU layer and a four-way softmax.

C. Two-Stage Training

Training follows the base paper (Table IV). In stage 1, all VGG16 weights are frozen and only the randomly initialized head is trained, so that its large early gradients do not disturb the pretrained features. In stage 2, the first ten VGG16 layers (blocks 1–3) remain frozen while blocks 4–5, which encode higher-level and more dataset-specific patterns, are fine-tuned together with the head at a ten-times smaller learning rate to avoid erasing pretrained knowledge.

D. Evaluation

We report accuracy, per-class recall (sensitivity) and specificity, macro one-vs-rest AUC, log loss on the full validation and test sets, the mean train–validation loss gap over the last five epochs, and expected calibration error (ECE, 15 bins) [18]. To audit leakage, we compute a 64-bit difference hash for every image and flag test images whose hash equals that of a training image. This gives a lower bound, since different slices of the same patient are not detected. The baseline was trained in two independent runs (the reproduction run and the ablation run) to check run-to-run variation; the Flatten–GAP comparison uses models trained in the same run. Following the base paper's 21 validation steps, validation during training uses only the first 672 of the 800 validation images; because these are ordered by class, they omit 128 pituitary images, so training-time validation loss is higher than the full-set values in Table VII, although it remains comparable between models.

VI. RESULTS

A. Reproduction of the Base Paper

Our reproduction run reaches 95.25% accuracy (762 of 800 test images), against the reported 99.24% (Table V). Fig. 4 shows

TABLE V
REPRODUCTION RESULTS ON THE TEST SET

Metric (%)	Wong et al. [1]	Ours
Accuracy	99.24	95.25
Macro sensitivity	99.18	95.25
Macro specificity	99.75	98.42

TABLE VI
TWO INDEPENDENT BASELINE (FLATTEN) RUNS

Run	Accuracy (%)	Glioma recall (%)	Unseen acc. (%)
Reproduction run	95.25	84.5	93.81
Ablation run	95.38	85.0	93.98
Difference	0.13	0.5	0.17

that three classes are within about one point of the original; the gap comes almost entirely from glioma, which accounts for 31 of our 38 test errors (17 predicted as meningioma, 12 as no tumor and 2 as pituitary). Fig. 3 shows typical glioma errors from the ablation run, analyzed in Section VII.

B. Test-Set Leakage

In our split, 202 of 800 test images (25.3%) share a hash with a training image. The reproduction model classifies 201 of these 202 correctly (99.50%), but only 561 of the remaining 598 (93.81%) (Fig. 5). Repeated images therefore raise measured accuracy by 1.44 points in our setting (95.25% versus 93.81%). The original split may be affected similarly but was not audited.

C. Baseline Stability Across Runs

Table VI compares the two independent baseline runs, trained with identical settings. Test accuracy differs by only 0.13 points (95.25% versus 95.38%), and glioma recall is 84.5% and 85.0%, confirming that the glioma weakness is systematic rather than due to chance.

D. Flatten Versus GAP

Table VII compares the two models trained in the same run under identical conditions. GAP is worse on almost every measure: test accuracy falls by 3.13 points, far more than the 0.13-point difference between the two baseline runs, and glioma recall falls by 12.5 points. Table VIII compares each outcome with its prediction: all five directional predictions failed, and the risk we flagged, a drop in pituitary recall, did not occur either (pituitary recall rose from 99% to 100%). Fig. 6 compares where the two models look.

E. Overfitting by Training Stage

The learning curves (Fig. 7) separate the two training stages, and Table IX summarizes them using the training-time validation subset (Section V-D). While VGG16 was frozen (stage 1), GAP behaved as hypothesized: its validation loss was 47% lower than Flatten's (0.68 versus 1.28) and its training loss stayed higher (0.15 versus 0.02), indicating far less memorization. However, its validation accuracy was also lower (85.9% versus 90.1%). Once blocks 4–5 were fine-tuned

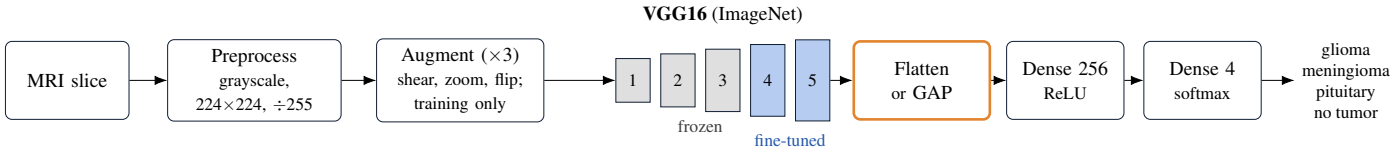


Fig. 2. Classification pipeline. Blocks 1–3 of VGG16 stay frozen throughout; blocks 4–5 are fine-tuned in the second training stage. The orange box is the only component that differs between the baseline (Flatten) and our model (GAP).

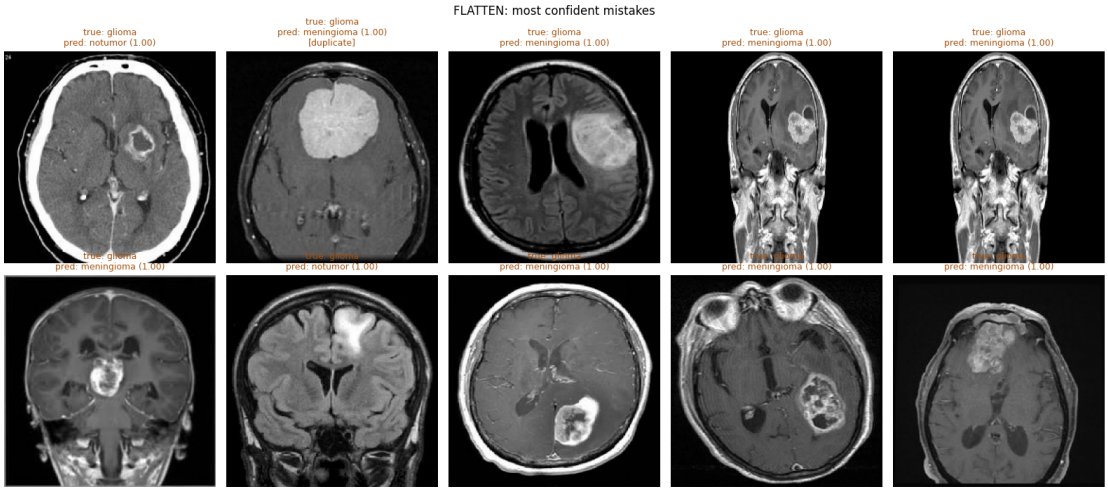


Fig. 3. The baseline's ten most confident test errors in the ablation run, all on gliomas: most are predicted as meningioma with near 100% confidence.

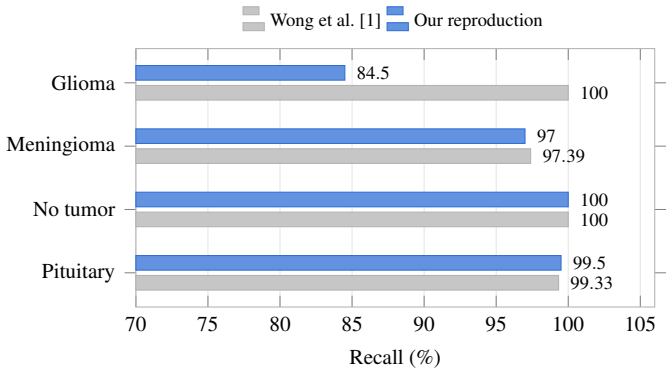


Fig. 4. Per-class recall of the original model and our reproduction. The axis starts at 70% so that differences are visible; the gap is concentrated in glioma.

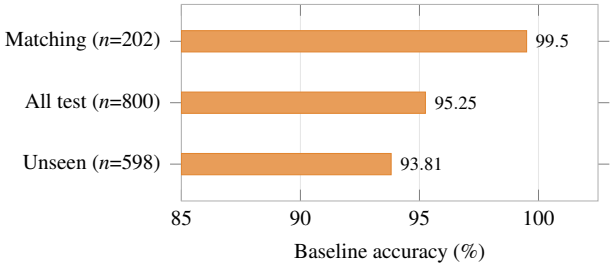


Fig. 5. Accuracy on test images that match a training image versus truly unseen images. A quarter of the test set is effectively already seen during training.

(stage 2), both models drove training loss towards zero and reached similar validation loss (about 1.0–1.1): the fine-tuned backbone, with about 13 million trainable weights, overfits regardless of the head.

VII. Discussion and Failure Analysis

A. Why the Hypothesis Failed

The hypothesis rested on two assumptions: that the classifier head was the main source of overfitting, and that position information was dispensable. Both proved wrong. The stage-wise curves show that the head *did* drive overfitting while the backbone was frozen, and GAP removed much of it, but fine-tuning 13 million backbone weights on a training set built from fixed augmented copies reintroduced overfitting that the head cannot control. Meanwhile, discarding spatial layout cost discriminative information: GAP's glioma errors rose from 30 to 55, spread across meningioma (26), no tumor (19) and pituitary (10). The anticipated pituitary risk did not occur (recall 100%), suggesting that pituitary tumors are recognized from appearance rather than position.

B. Where Do the Models Fail?

Fig. 8 shows that glioma is the dominant error source for both models, consistent with the persistent confusion among tumor types reported in the literature (G2) [6]. In the ablation run's baseline, 19 of 30 glioma errors are predicted as meningioma and 10 as no tumor. The baseline's most confident mistakes (Fig. 3) are all made with near 100%

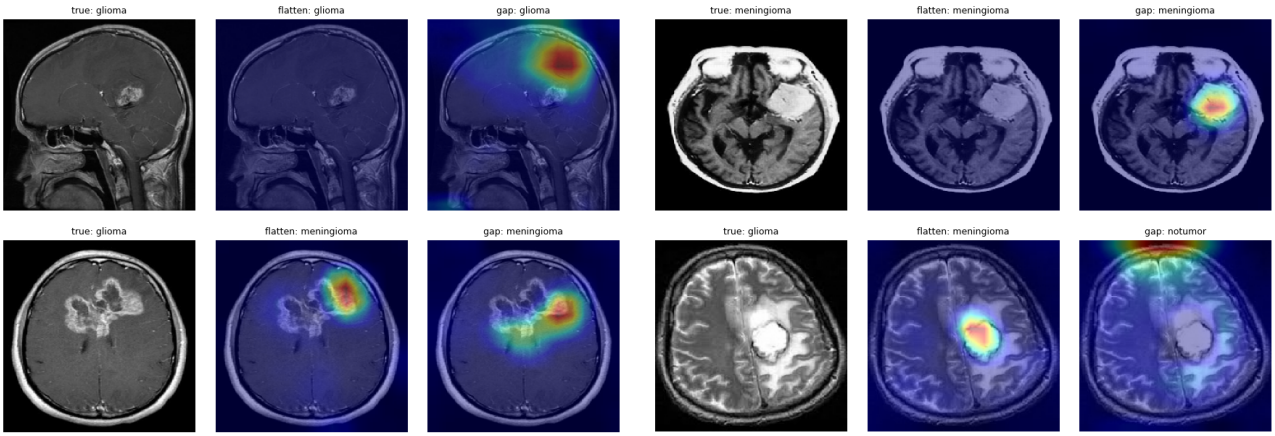


Fig. 6. Grad-CAM heatmaps. Top: a correctly classified glioma (left) and meningioma (right). Bottom: two misclassified gliomas. Each triple shows the input, the Flatten model and the GAP model.

TABLE VII
SINGLE-VARIABLE ABLATION (SAME RUN, TEST SET UNLESS NOTED)

Metric	Flatten	GAP
Test accuracy (%)	**95.38**	92.25
Macro AUC	**0.989**	0.986
Test loss	**0.450**	0.755
Validation loss (full set)	**0.616**	0.899
Train–val loss gap (last 5 epochs)	**1.061**	1.090
Recall: glioma (%)	**85.0**	72.5
Recall: meningioma (%)	**97.5**	96.5
Recall: no tumor (%)	100.0	100.0
Recall: pituitary (%)	99.0	**100.0**
Macro specificity (%)	**98.46**	97.42
ECE	**0.037**	0.056
Accuracy, unseen images (%)	**93.98**	89.80
Accuracy, duplicate images (%)	99.50	99.50

TABLE VIII
PREDICTIONS VERSUS OBSERVED OUTCOMES

Measure	Predicted	Observed	Held?
Test accuracy	$+1$ to $+3$ pts	-3.13 pts	no
Validation loss	30–60% lower	46% higher	no
Train–val gap	smaller	$1.06 \rightarrow 1.09$	no
Glioma recall	higher	$85.0 \rightarrow 72.5\%$	no
Pituitary recall	may decrease	$99 \rightarrow 100\%$	risk avoided
Macro AUC	slightly higher	$0.989 \rightarrow 0.986$	no

confidence and share recurring traits: large, sharply bounded masses that resemble meningiomas, varied MRI sequences and orientations, and at least one image that resembles a CT rather than an MRI scan. The test set also contains repeated glioma images. Grad-CAM maps (Fig. 6) are often nearly empty for the Flatten model, consistent with saturated, over-confident softmax outputs, whereas GAP's maps sometimes highlight the tumor but in misclassified gliomas also focus on the skull boundary. As noted in G1, this assessment is qualitative.

C. Why the Reproduction Gap?

Our accuracy is 3.99 points below the original (95.25% versus 99.24%). Likely contributors are a different dataset version with a different test set, unreported augmentation strengths, random initialization and GPU non-determinism, and leakage whose size in the original split is unknown. Without the original code, these factors cannot be separated.

TABLE IX
MEAN OF THE LAST 10 EPOCHS OF EACH TRAINING STAGE (TRAINING-TIME VALIDATION SUBSET)

	Stage 1 (frozen)		Stage 2 (fine-tuned)	
	Flatten	GAP	Flatten	GAP
Training loss	0.022	0.150	0.011	0.017
Validation loss	1.284	**0.679**	1.029	1.100
Validation accuracy (%)	**90.1**	85.9	93.3	92.6

D. A Follow-Up Hypothesis

Offline augmentation creates two fixed copies of each image, so every epoch draws its batches from the same fixed set of 16,800 images, which the fine-tuned network sees repeatedly and can memorize. We therefore propose, as future work, that generating fresh random augmentations in every epoch (online augmentation) will reduce stage-2 overfitting and improve glioma recall, and that keeping the checkpoint with the lowest validation loss will avoid the late-training rise in validation loss visible in Fig. 7.

VIII. LIMITATIONS

This study uses a single public dataset with no external validation, so performance on scans from other hospitals is unknown. GAP was trained only once, and two baseline runs give only a rough estimate of variation, although the 3.13-point drop is far larger than the 0.13-point difference between them. The duplicate audit misses different slices of the same patient, and without patient identifiers a patient-level split was not possible. Leaked images were measured but not removed. The models classify single 2D slices, and the dataset mixes MRI sequences and contains at least one image that appears to be CT. **No clinical claim is made.**

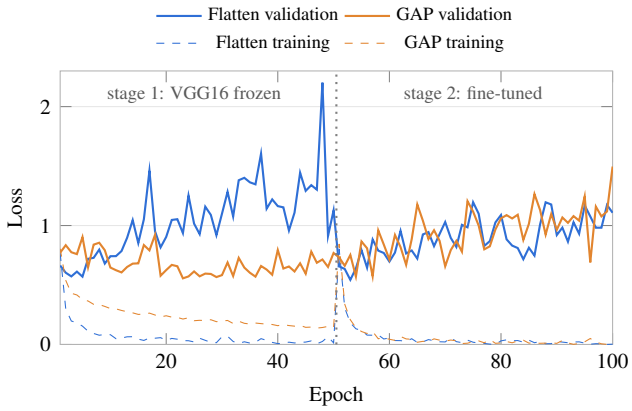


Fig. 7. Training (dashed) and validation (solid) loss. GAP keeps validation loss low while the backbone is frozen, but after fine-tuning starts (dotted line) both models overfit to a similar degree. Validation loss is measured on the training-time subset (Section V-D).

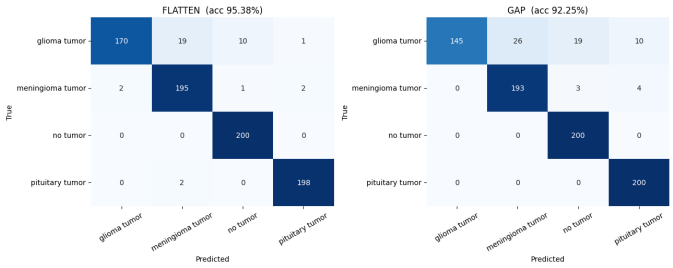


Fig. 8. Test-set confusion matrices for Flatten (left) and GAP (right).

IX. Conclusion

We rebuilt a VGG16 brain tumor classifier from its published description and reached 95.25% accuracy (95.38% in a second run), against the reported 99.24%, with the difference concentrated in glioma. A quarter of the test images match training images, so accuracy on truly unseen images is lower than the headline figure suggests. Our hypothesis, stated in advance, that a global-average-pooling head would reduce overfitting and raise accuracy, was refuted: GAP reduced overfitting only while the backbone was frozen and lowered accuracy overall, mainly by losing spatial information needed for glioma. The failure analysis shows that overfitting arises chiefly during fine-tuning, which motivates online augmentation as the next hypothesis. Future work should also use patient-level splits, multiple seeds, external test data and quantitative heatmap evaluation.

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